BY-COVID
beyond covid

How to make your sensitive data FAIR, Challenges and use-cases session: Use-case in social sciences

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Social science data

- What kind of data are collected is mainly determined by the research questions
- Social science research data are typically questionnaire surveys, interviews, focus group discussions, written material, visit or meeting recordings, official documents, archival material, websites, register data or social media data
  - Quantitative data can be collected through interviews, postal or online questionnaires, by using existing source material, or by measuring
  - Qualitative data are often collected by recording individual interviews, group interviews, sessions or meetings as audio or video files; written material collection is usually initiated by publishing writing requests or invitations, and then collecting the writings via email, post or a specifically created website
- In many cases, datasets are produced by one researcher or a small team of researchers

This presentation's focus is on data collected by researchers using questionnaires and/or interviews.

FAIR principles

Source: https://www.go-fair.org/fair-principles/
See also: Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016). https://doi.org/10.1038/sdata.2016.18

F1: (Meta)data are assigned globally unique and persistent identifiers
F2: Data are described with rich metadata
F3: Metadata clearly and explicitly include the identifier of the data they describe
F4: (Meta)data are registered or indexed in a searchable resource

A1: (Meta)data are retrievable by their identifier using a standardised communication protocol
- A1.1: The protocol is open, free and universally implementable
- A1.2: The protocol allows for an authentication and authorisation procedure where necessary

A2: Metadata should be accessible even when the data is no longer available

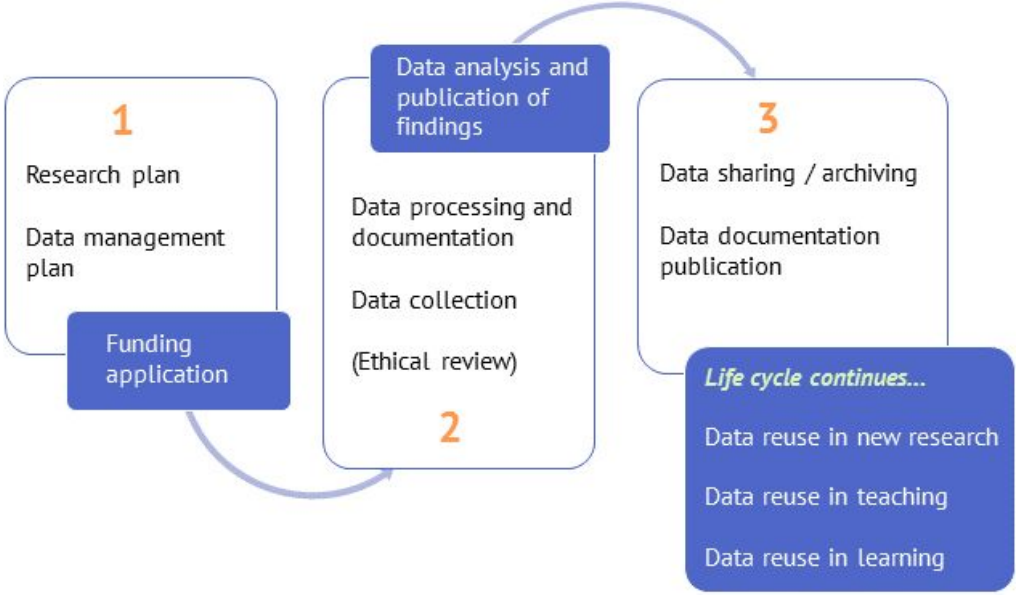
I1: (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation
I2: (Meta)data use vocabularies that follow the FAIR principles
I3: (Meta)data include qualified references to other (meta)data

R1: (Meta)data are richly described with a plurality of accurate and relevant attributes
- R1.1: (Meta)data are released with a clear and accessible data usage license
- R1.2: (Meta)data are associated with detailed provenance
- R1.3: (Meta)data meet domain-relevant community standards

Good research data management

- Making (meta)data FAIR starts with good data management
- Data and related metadata need to be created, preserved and organised in a way that ensures that data remain accessible, understandable and reliable
- Data protection and security must be maintained over the whole data life cycle

=> responsibly and ethically FAIR (meta)data



Picture source: https://www.fsd.tuni.fi/en/services/data-management-guidelines/why-are-research-data-managed-and-reused/

Sharing sensitive data

- Especially for (sensitive) personal data, there may be a perceived tension between data sharing and data protection.
- Sensitive data can be shared
  - Combination of gaining consent, anonymising data, gaining clarity over who owns the copyright to data, and controlling access to data can enable the ethical and legal sharing of data.
- In social sciences, data are typically collected from individuals so informed consent is important
  - The research participants must be informed about storage, archiving and reuse of the research data
- GDPR does not apply to anonymized data
  - Researchers sometimes reluctant to anonymize
  - Anonymization can be resource-intensive

Sensitive data and FAIR

- FAIR data **is not** the same as “open and free”
- Sensitive data can be made FAIR by making its metadata FAIR
- Even/especially if data access is restricted, metadata should be made openly available
  - General-level metadata is anonymous (except for bibliographical information about data creators)
  - In social sciences, variable-level metadata is usually no risk for identification
  - The more metadata available, the easier it is for a potential re-user to assess if data is suitable for their purposes, without needing access to actual data
- Important to follow community best practices

FAIR data and metadata

- Both data and metadata should be made as FAIR as possible
  - Can be resource-intensive
  - Social science research teams tend to be small and don't have a data steward
- Current FAIR assessment tools usually assess the FAIRness of metadata
  - 'FAIR enough' is not well-defined
  - Especially difficult to know or assess when data is FAIR enough
- Anyone can test metadata records using e.g. F-UJI (https://www.f-uji.net/)
  - Just enter an identifier (e.g. DOI, URL) of the data you wish to assess
  - F-UJI includes social science metrics
- Use FAIR and assessments to assist in improving data management!!

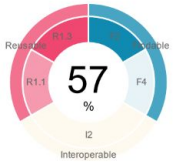
Evaluated Resource:

Parties and Local Democracy 1996: Party Members

Save | {JSON} | New

FAIR level:	initial
Resource PID/URL:	https://datacatalogue.cessda.eu/detail?lang=en&q=3c76a859c8744280887c39d9fb46265d22079dcd619f25e5f328f122fba596c0
DataCite support:	enabled
Metric Version:	metrics_v0.5ss
Metric Specification:	https://doi.org/10.5281/zenodo.6461229
Software version:	3.1.0
Download assessment results:	{JSON}
Save and share assessment results:	

Summary:



	Score earned:	Fair level:
Findable:	2 of 3	moderate
Accessible:	Not tested in this metrics	
Interoperable:	0 of 1	incomplete
Reusable:	2 of 3	moderate

Results:

urce:

lisdemokratia 1996: puoluejäsenet

Save | {JSON} | New

	moderate
	https://services.fsd.tuni.fi/catalogue/FSD1022
	enabled
	metrics_v0.5
	https://doi.org/10.5281/zenodo.6461229
	3.1.0
Download assessment results:	{JSON}
Save and share assessment results:	

Summary:

	Score earned:	Fair level:
Findable:	6 of 7	advanced
Accessible:	2 of 3	moderate
Interoperable:	1 of 4	initial
Reusable:	3 of 10	initial

FAIR + time

- High-quality data have the potential to be reused in many ways now and in the future.
- Archiving and publishing data properly is at the core of making data FAIR
    - Publishing sensitive data = publishing metadata about the data
- Sharing data via a trusted repository (e.g. a certified TDR) ensures that data stays FAIR over time; in social sciences, use CESSDA data archives
- TDRs usually also support data curation and making (meta)data FAIR, and enable metadata harvesting/sharing (e.g. from CDC to COVID-19 Data Portal)
- Residual risk assessment - a good repository will re-assess the robustness of the anonymisation periodically

FAIR metadata basics

- If you are a researcher, open your data via a repository that enables and supports FAIR + time, has expertise on sharing sensitive data, and can support and help you in all below!

- Data needs to have a persistent identifier (PID) and the PID needs to be included in the metadata
- Core descriptive metadata needs to be provided in a common metadata standard (like Dublin Core, schema.org, DCAT, Datacite…)
- Data access conditions need to be described and a link to data use license included in the metadata
- Metadata needs to be embedded in the landing page, linked via typed links (FAIR signposting)
- Metadata needs to be structured using a machine-actionable language
- Data should be documented using community specific metadata standard(s)
    - in social sciences: DDI standards
- Metadata should include vocabulary namespace URIs
- Metadata should include machine-actionable references to other (meta)data
- Metadata should include provenance information

(Based on: Nordling, J., Mihai, H., Meerman, B., Alaterä, T., Kleemola, M., & Livenson, I. (2022). D4.3 Report on Nordic and Baltic repositories and their uptake of FAIR. Zenodo. https://doi.org/10.5281/zenodo.6880904)

Sources, further reading

- https://dmeg.cessda.eu/Data-Management-Expert-Guide/
- https://www.fsd.tuni.fi/en/services/data-management-guidelines/

Data Management Expert Guide

cessda

Data Management Expert Guide (DMEG)

The DMEG is designed by European experts to help social science researchers make their research data Findable, Accessible, Interoperable and Reusable (**FAIR**).

You will be guided by different European experts who are - on a daily basis - busy ensuring long-term access to valuable social science datasets, available for discovery and reuse at one of the CESSDA social science data archives.

You can download the full DMEG for your personal study offline (DOI: 10.5281/zenodo.3820473). PDFs for every single chapter are also available for being printed as handouts for training.

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